

4. All terrestrial plants lose water from their leaves by a process called transpiration.

(a) A student used a potometer to estimate the transpiration rate of a leafy shoot over a twenty-four hour period.

(i) Explain why the water uptake from the potometer gives an estimate of the transpiration rate and not its true value. [1]

.....

.....

(ii) Give **three** other variables that would need to be controlled when using a potometer to investigate the difference in transpiration rates of leafy shoots of **two** different plant species at the same light intensity and temperature. [3]

.....

.....

.....

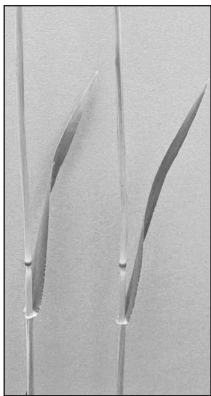
(b) The table shows the average density and distribution of stomata on the leaves of oak and wheat.

Plant	Density of Stomata (number mm ⁻²)	
	Upper leaf surface	Lower leaf surface
Wheat	50	40
Oak	0	340

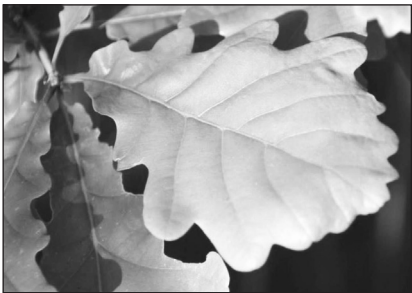
Explain the distribution of stomata in wheat and oak. [2]

.....

.....



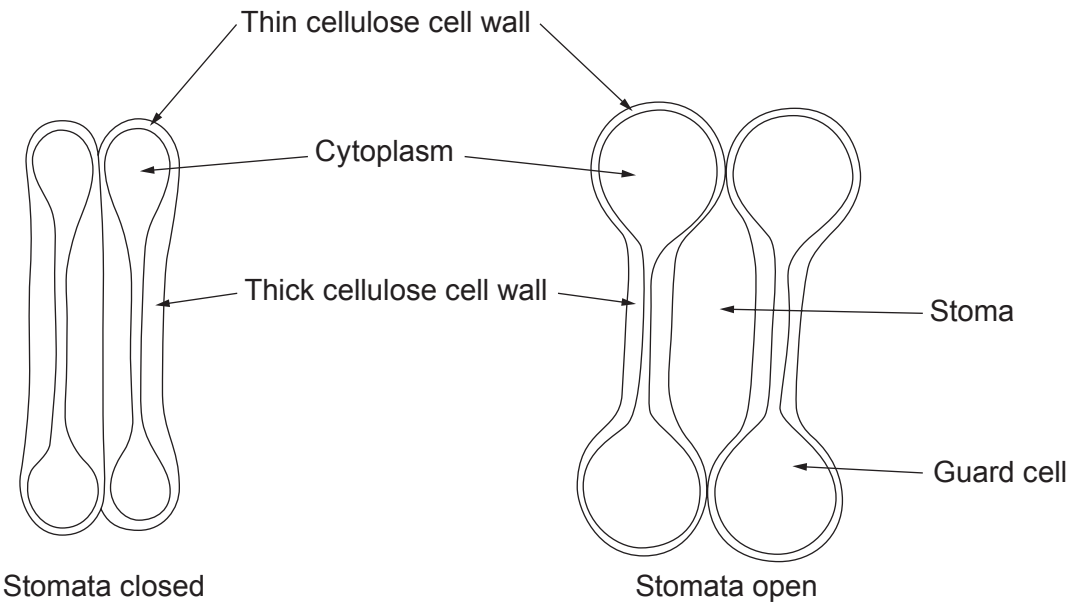
Wheat (*Triticum aestivum*)



Oak (*Quercus robur*)



(c) The diagrams below show the arrangement of guard cells in a leaf of wheat under two different environmental conditions.



Stomata open when potassium ions are actively transported into the guard cells. Explain how this would cause wheat stomata to open. [5]

.....

.....

.....

.....

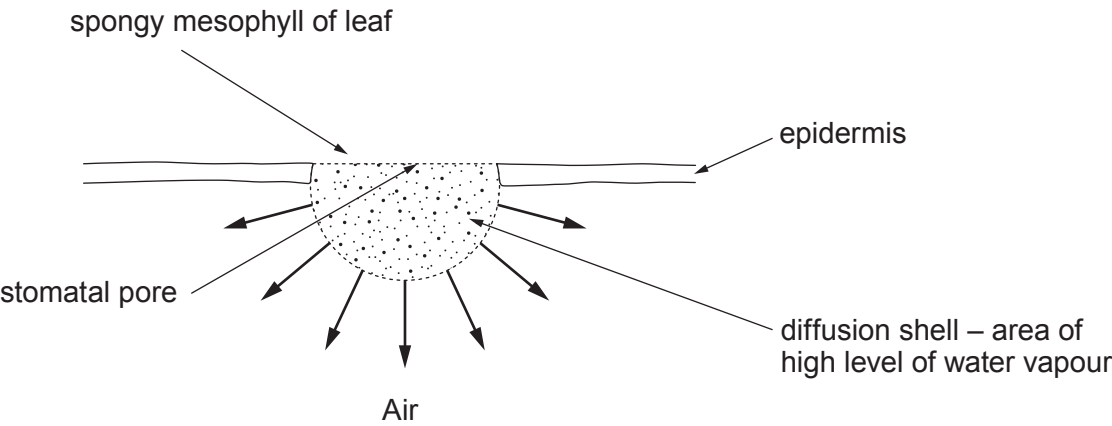
.....

.....

.....



(d) Water is lost through stomata by transpiration. This results in a shell of water vapour forming over the stomatal pores as shown.



Water molecules from the edges of stomatal pores evaporate more readily into the air.

The stomatal density and the diameter of the stomatal pores were recorded for the leaves of two species of plant. The stomatal pores were circular and the total area of the pores was approximately the same in both plants.

Plant species	Density of stomata (number mm ⁻²)	Mean diameter of stomatal pore (µm)	Total circumference of stomata (µm mm ⁻²)
A	110	8	2760
B	423	4	

Circumference of a circle = $2\pi r$
 r = radius of circle
 π = 3.14

(i) Calculate the total circumference of the stomatal pores for species **B**. Give your answer to three significant figures. [2]

Circumference =

(ii) Use these figures to identify which plant species, **A** or **B**, has the higher transpiration rate. Explain your answer. [1]

.....
.....

